

cohort will shed light on the long-term sex-specific impact of APOE genotype on brain structure and cognition in preclinical populations with risk for AD.

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Poster No 204

Frontoparietal connectivity strength relates to changes in disability metrics in multiple sclerosis

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Introduction: Disease progression is variable in multiple sclerosis (MS). MRI-based measures that track and predict MS disease progression could identify patients who are at risk of decline and serve as outcome measures in clinical trials of novel disease modifying treatments. Here, we focus on within-network resting state functional connectivity (rsfMRI) of the frontoparietal network (FPN), considered critical for coordination of brain function and cognitive control¹. We assess connectivity of the FPN in people with MS (pwMS) at two time points, hypothesizing that changes in rsfMRI strength will be related to measures of disability, particularly to cognitive function.

Methods: Under an IRB-approved protocol, 47 patients with MS [mean age: 50.2 ± 8.5, 10 males, median EDSS: 3.5, range 1.5-6.5] completed an MRI, clinical evaluation, and cognitive testing at two time points separated by one year. Measures included the Multiple Sclerosis Functional Composite (MSFC) and tests of memory, processing speed, and executive function. A whole-brain T1-weighted MP2RAGE (0.75mm³) and rsfMRI scan were acquired on a Siemens 7T Magnetom with a SC72 gradient (Siemens Medical Solutions, Erlangen) using a 1-Tx and 32-Rx channel head coil (Nova Medical). RsfMRI acquisition parameters were: 132 repetitions of 81 1.5mm thick axial slices acquired with TE/TR=21ms/2800ms, voxel size 0.75×0.75×1.5mm³, matrix 160×160, FOV 210mm×210mm, receive bandwidth = 1562 Hz/pixel, eyes closed. All rsfMRI scans were corrected for motion and physiologic noise, detrended, and lowpass filtered^{2,3}. Freesurfer 7.1 was used to generate cortical grey matter parcellations and apply the Yeo 7-network FPN template⁴. The MP2RAGE was coregistered and warped to rsfMRI and rsfMRI volumes from visit 1 (V1) were coregistered to visit 2 (V2) for each participant (AFNI). Using a previously described method⁵ and the FPN template, the V1 rsfMRI scan was used to identify 9-voxel in-plane seeds in the bilateral grey matter of the middle frontal gyrus (MFG; BA 9), representing the FPN. Seeds were propagated to V2 for each participant and used to calculate whole-brain normalized⁶ connectivity maps at both visits. Individual FPN maps were moved to Talairach space⁷. FPN connectivity was averaged across all participants (Figure 1) and used to create a mask of significant regions. AFNI 3dttest++ was used to perform a voxel-wise t-test of masked rsfMRI between V1 and V2. In regions showing significant V1-V2 differences, mean rsfMRI strength and the change in rsfMRI was correlated with performance on clinical and cognitive measures.

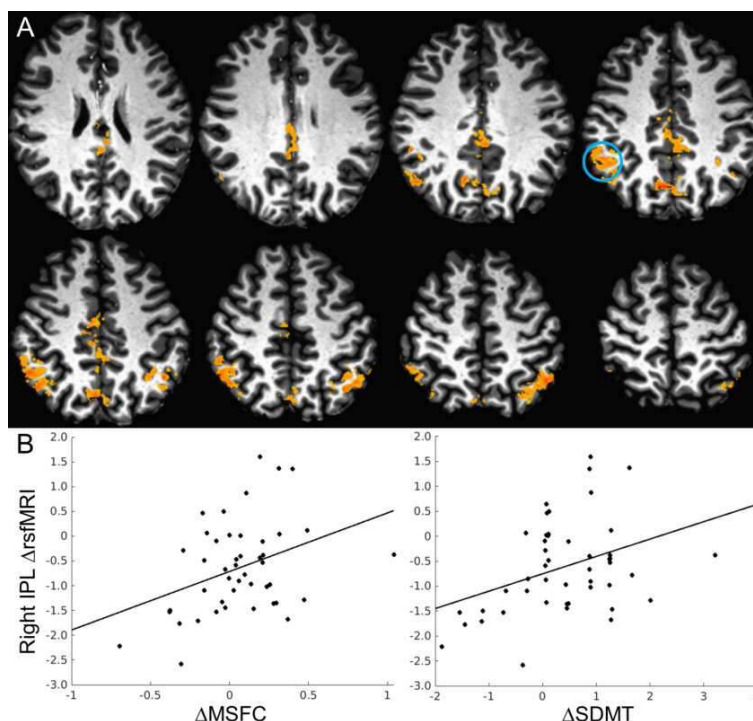


Figure 2. A. V1-V2 differences in FPN rsfMRI ($p < 0.05$, cluster size 1500). All regions show stronger connectivity at V1. B. The V1-V2 change in rsfMRI of the right IPL (circled in blue) is related to the V1-V2 change in MSFC ($r = 0.37$, $p < 0.012$) and SDMT ($r = 0.369$, $p < 0.011$).

Results: All significant V1-V2 differences were the result of a decline in rsfMRI at V2 (Figure 2). V1 rsfMRI was positively related to V1 MSFC, the symbol digit modalities test (SDMT), and memory ($p < 0.05$), but did not survive a correction for multiple comparisons. The V1-V2 change in rsfMRI of the left precuneus and right inferior parietal lobule IPL was positively related to change on the MSFC ($r = 0.45$, $p < 0.002$; $r = 0.37$, $p < 0.012$, respectively) and the SDMT ($r = 0.53$, $p < 1.1 \times 10^{-4}$; $r = 0.369$, $p < 0.011$, respectively), so that a decline in functionality was associated with a decline in rsfMRI (Figure 2).

Conclusions: Here, we show that the change in rsfMRI of the FPN is related to the change in MSFC, driven by performance on the SDMT. Although the direction of the relationship between behavior and V1 rsfMRI was in line with our previous findings⁸, these relationships did not reach significance. SDMT performance draws on multiple cognitive domains, including speed of processing, attention, and memory. Degradation of the FPN may lead to a breakdown of coordination between networks responsible for these functions. Future work will investigate this possibility by assessing between-network connectivity. This work was supported by the Department of Defense (MS150097). The authors acknowledge technical support by Siemens Medical Solutions.

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